

STANDARD OPERATING PROCEDURE FOR OPERATION for DRINKING RO PLANT.

PPE: Use hand gloves, Safety helmet and Safety shoes.

Startup procedure:

- Ensure all valves are in the correct position.
- Check that the raw water tank is adequately filled.
- Start the raw water pump, then start the dosing pumps for chemicals.
- Flush the sand and carbon filters, then switch them to filter mode.
- Vent air from the pre-filter housings.
- Turn on the high-pressure (HP) pump and gradually adjust the rejection valve to control the flow and pressure.
- Run the system to drain for 30 minutes to an hour after startup to flush the system.

Shutdown procedure:

- Turn off the HP pump.
- Flush the RO membranes with clean permeate water for 15–20 minutes to prevent membrane scaling.
- Turn off the raw water pump.
- Close all necessary valves.

Daily log and maintenance:

- Record system pressures (inlet and outlet), flow rates, and water quality parameters (TDS, pH).
- Backwash the sand and carbon filters to remove trapped solids.
- Check chemical dosing tanks and top up as needed.
- Monitor pressure gauges on pre-filters to identify cartridge filter replacement needs.

Periodic maintenance:

- **Membrane cleaning (CIP):** If production rates drop or water quality declines, use a Clean-In-Place (CIP) system to circulate chemical cleaners through the membranes.

- **Replacement of consumables:** Replace filter cartridges, antiscalant chemicals, and activated carbon media on a recommended schedule.

Quality control standards

The final drinking water must meet regulatory standards for safety.

- **Parameter monitoring:** Regularly check the water's pH, TDS, and microbiological content.
- **TDS levels:** Monitor the final product water's TDS level, which should be within safe drinking water limits. For example, standards suggest TDS to be between 100 and 500 ppm.
- **National standards:** Adhere to national drinking water regulations, such as those from the EPA (in the US) or the Central Pollution Control Board (in India).
- **International standards:** Follow global guidelines from organizations like the World Health Organization (WHO), especially for international operations.

SOP NO.: PIL/PTC/RO/06 REV.00 DATE: 30.04.2025.